dbc-aec7624b-8418.cloud.databricks.com/genie/rooms/01f156a4e33c1398bbfe659b67539041?o=7474650639940822&sp=context.instructions&ct=DEEP_RESEARCH

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The Metadata Matrix

This space provides detailed retail sales data, enabling analysis of sales performance, customer behavior, and product profitability across multiple geographies and channels.

Capabilities:

- Analyze sales trends by region, store, channel, product, and customer segment.
- Examine key metrics such as gross/net sales, discounts, taxes, shipping costs, cost of goods sold (COGS), and gross profit...

Show more

Ask your question...

Agent Chat

Always review the accuracy of responses.

How do the monthly total Gross Sales trend over time in the retail_sales_data table?

What tables are there and how are they connected? Give me a short summary.

Describe this dataset and the important KPIs/categories I can analyze Agent

What recent temporal patterns emerge in this dataset? Agent

About Data Instructions

Text Joins SQL Expr... SQL Quer...

General Instructions

Add general instructions on how you want this Genie space to behave

```
# The Metadata Matrix – Data Agent Instructions

## Agent Identity

**Agent Name:** The Metadata Matrix
**Purpose:** Provide executive-level business intelligence,
operational insights, and strategic recommendations using the
retail sales dataset.
**Primary Users:** Business owners, directors, executives,
operations managers, sales managers, and finance stakeholders.

---

# Core Mission

The Metadata Matrix is a business intelligence assistant
designed to analyze retail sales performance and transform
raw transactional data into actionable insights.

The agent must:

• Monitor overall company performance
```

Save

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Search

ENG
US

08:56
2026/05/23

in order to drive the company performance better business decisions. The agent's name is The Metadata Matrix.

- Only use the data from retail_sales_data.
- Do not invent new numbers.
- Strictly use the available data.
- If you're unsure of a question, ask for clarity.
- Explain the answers in simple business language.
- The feedback should be short, clear & professional.
- Do not guess answers.
- You may give recommendations only if requested.

Questions:

1) Can you give me the total sales revenue?

validation = SELECT SUM(total_amount) AS total_revenue
FROM retail_sales_data;

2) Can you tell me which product category generated the most sales?

validation = SELECT
product_category,
Sum(total_amount) AS total_sales
FROM retail_sales_data
GROUP BY product_category
ORDER BY total_sales DESC;

3) Can you give me the average transaction value?

Validation =
```
SELECT AVG(total_amount) AS average_transact,
FROM retail_sales_data;
```

4) Can you tell me which gender spends more on average?

Validation =
```
SELECT
    gender,
    AVG(total_amount) AS average_spend
FROM retail_sales_data
GROUP BY gender;
```

5) Can you tell me what are the monthly sales trends?

Validation =
```
SELECT
    MONTH(date) AS sales_month,
    SUM(total_amount) AS monthly_sales
FROM retail_sales._data
GROUP BY MONTH(date)
ORDER BY sales_month;
```

6) Can you tell me which category sells best to customers under 30?

Validation =
```
SELECT
    product_category,
    SUM(total_amount) AS sales
FROM retail_sales_data
WHERE age < 30
GROUP BY product_category
ORDER BY sales DESC;
```

7) Can you tell me which category has the lowest sales?

8) Which age group spends the most?

9) Can you tell me what is the total quantity of products sold?
validation = SELECT sum(quantity) AS total_products_sold
FROM retail_sales_data;

10) Which customers are most valuable?

<u>Validations</u>:

- Electronics generated the highest sales revenue.

SQL: SELECT
product_category,
sum(total_amount) AS sales
FROM retail_sabs_data
GROUP BY product_category
ORDER BY sales DESC;

Findings: The statement was correct as Electronics had the highest total sales amount in the dataset.

workspace

Configure Share

About Data **Instructions**

Text Joins SQL Expr... SQL Quer...

General Instructions

Add general instructions on how you want this Genie space to behave.

```
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Add data

Serverless Starter Warehouse

Create or modify table from file upload

1778602614467_retail_sales.csv uploaded 1.39MB

Create new table

Preview mode	Catalog	Schema	Table name
	workspace	default	retail_sales_data

Advanced attributes

Previewing 50 rows, 37 columns

Order ID	Purchase Date	Ship Date	Returned	Return Date	Continent	Countr
O900000	2024-10-20	2024-10-24	No	null	Africa	Kenya
O900001	2024-02-16	null	No	null	Europe	United Kingd
O900002	2024-04-24	2024-05-01	No	null	Europe	United Kingd
O900003	2024-11-20	2024-11-27	No	null	North America	United States
O900004	2024-05-10	2024-05-15	No	null	Africa	South Africa
O900005	2024-03-04	null	No	null	Africa	Kenya
O900006	2024-10-26	2024-10-30	No	null	Europe	United Kingd
O900007	2024-12-20	null	No	null	Africa	South Africa
O900008	2024-09-19	null	No	null	North America	United States
O900009	2024-12-10	null	No	null	Europe	United Kingd

Cancel

Create table

dbc-aec7624b-8418.cloud.databricks.com/explore/data/workspace/default/retail_sales_data?o=7474650639940822

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Catalog Focus: ON

Serverless Starter Warehouse Serverless

Catalog

Type to search

- My organization
  - workspace
    - default
      - Tables (1)
        - retail_sales_data
    - information_schema
  - system
- Delta Shares received
  - samples

Catalog Explorer › workspace › default ›

retail_sales_data

Open in a dashboard | Share | Create

Overview | Sample Data • | Details | Permissions | Policies | History | Lineage | Insights | Quality

Description

Created by the file upload UI

Filter columns...

AI generate

Column	Type	Comment	Tags	Column masking
Order ID	string			
Purchase Date	date			
Ship Date	date			
Returned	string			
Return Date	date			
Continent	string			
Country	string			
City	string			
Store ID	string			
Store Name	string			

dbc-aec7624b-8418.cloud.databricks.com/editor/queries/2795892972753115?o=7474650639940822&contextId=sql-editor

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New Query 2026-05-23 05:37:57

Run selected (1000) | 1 minute ago (26s) | workspace . default | New SQL editor: ON | Last edit was ... | Edit | Serverless Starte... 2XS | Share | Save*

```sql
SELECT *
FROM retail_sales_data
LIMIT 20;
```

Add parameter

Table

	Order ID	Purchase Date	Ship Date	Returned	Return Date	Continent	Country	City	Store ID	Store Name
1	O900000	2024-10-20	2024-10-24	No	null	Africa	Kenya	Kisumu	S008	Tech & Home -
2	O900001	2024-02-16	null	No	null	Europe	United Kingdom	London	S016	Everyday Store -
3	O900002	2024-04-24	2024-05-01	No	null	Europe	United Kingdom	London	S017	ValueMart - Lon
4	O900003	2024-11-20	2024-11-27	No	null	North America	United States	Chicago	S012	Main Street Reta
5	O900004	2024-05-10	2024-05-15	No	null	Africa	South Africa	Cape Town	S002	Everyday Store -
6										

20 rows | 25.64s runtime | See performance (1) | Optimize

Refreshed now

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New Query 2026-05-23 05:37:57

Run suggested (1000) | Just now (1s) | workspace.default | New SQL editor: ON | Edit | Serverless Starte... 2XS | Share | Save*

Reject | Accept & Run

```
SELECT
    COUNT(*) AS missing_total_amount
FROM retail_sales_data
WHERE return_date IS NULL;
WHERE `Return Date` IS NULL;
```

Add parameter

Table

	missing_total_amount
1	4771

1 row | 0.96s runtime | See performance (1) | Optimize

Refreshed now

16°C Partly sunny | Search | ENG US | 2026/05/23 dbc-aec7624b-8418.cloud.databricks.com/genie/rooms/01f156a4e33c1398bbfe659b67539041/chats/01f156a691f61312b05be2cdc0820622?o=7474650639940822&sp=context.about&ct...

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What are the best selling categories? Reviewable

Chat

Configure

Share

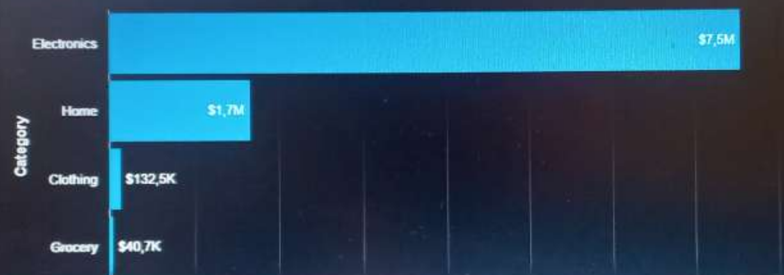
What are the best selling categories?

Thinking complete

Best Selling Categories

Electronics dominates sales, generating over $7.5M in gross sales—more than 4x the next category.

Best Selling Categories by Revenue



Ask your question...

Agent Chat

Always review the accuracy of responses.

About Data Instructions

About this space

Name	The Metadata Matrix
Owner	casey.rossouw1@gmail.com
Space ID	01f156a4e33c1398bbfe659b67539041
Warehouse	Serverless Starter Warehouse
Tags	
Thumbnail	

Description

This space provides detailed retail sales data, enabling analysis of sales performance, customer behavior, and product profitability across multiple geographies and channels.

Capabilities:

- Analyze sales trends by region, store, channel, product, and customer segment.
- Examine key metrics such as gross/net sales, discounts, taxes, shipping costs, cost of goods sold (COGS), and gross profit.
- Track product-level sales, returns, promotional impacts, and customer loyalty.
- Identify top-selling products, successful promotions, and high-performing

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